Automated Retail Analytics

Sample Report — Retail Sales & Customer Analysis

Data Source

E-commerce Sales & Customer Data (2009–2011)

Analyses Included

① Data Cleaning & Quality Assurance

② Sales Trend Analysis (Monthly / Weekly / YoY)

③ Product Performance & Pareto Analysis

④ Customer Segmentation (RFM)

Report Date: October 19, 2026

Author: Senior Data Scientist

Available for freelance data analysis engagements

This is a sample report. Fully customised to your business data and goals.

Background & Objective

An e-commerce retailer held two years of sales and customer data but lacked the tools to turn it into decisions. This report structures the data across three lenses — revenue trends, product performance, and customer behaviour — and surfaces next actions.

£17.7M	5,878	Top 20%
Total Revenue	Customers Analysed	of products drive 80%
2009–2011	After data cleaning	of revenue (Pareto)

Key Findings (3)

1. Revenue shows strong November seasonality, consistent across both years.
   → Stock and staffing must be prepared by mid-September at the latest.

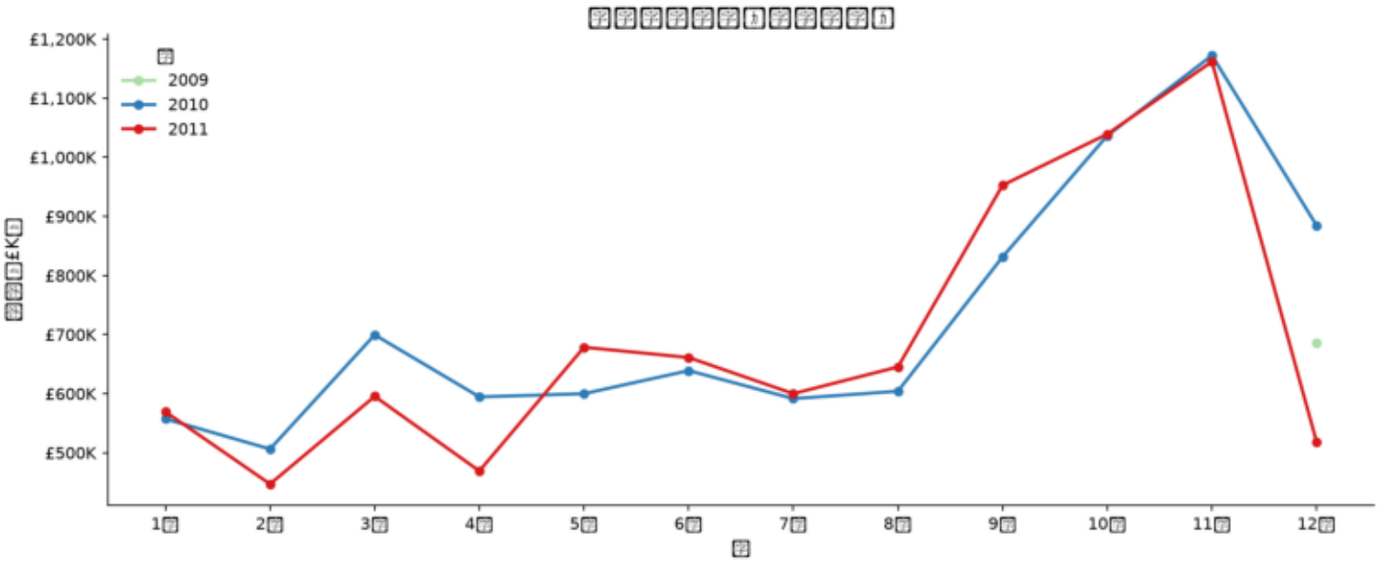
2. The top 20% of products generate the majority of revenue (Pareto principle).
   → Preventing stockouts on hero SKUs is the highest-leverage inventory action.

3. Champion customers are a small share of the base but contribute a disproportionate share of revenue. Retaining them delivers the best return on any retention spend.

Dataset Snapshot

Analysis Period	December 2009 – December 2011
Transaction rows	˜805,000 (after cleaning)
Unique customers	5,878
Unique products	5,283 SKUs
Markets	UK-centric · 40+ countries

Monthly Revenue — Year-on-Year Comparison



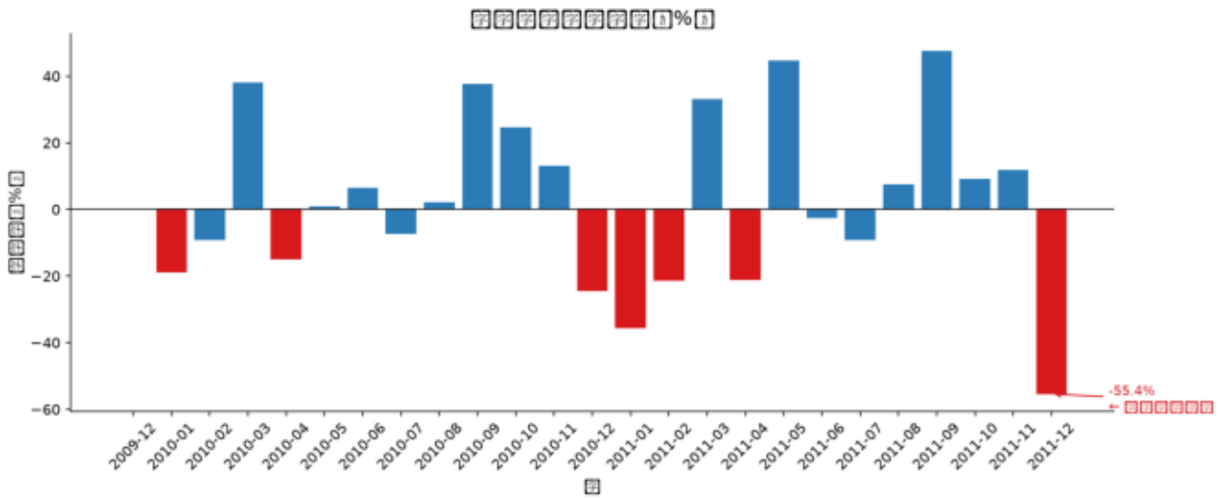
Analysis Commentary

- A November revenue spike is confirmed for two consecutive years, driven by holiday-season demand.

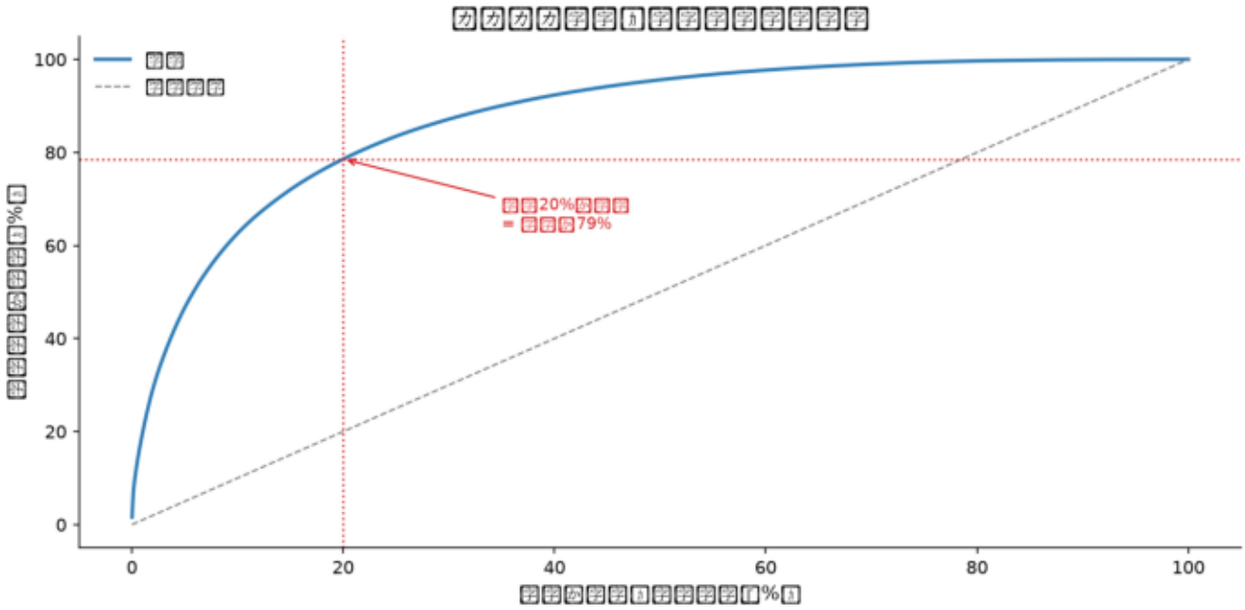
- Overall revenue grew year-on-year in 2011, confirming the underlying business growth trajectory.

→ Inventory and staffing prep must begin by mid-September at the latest. Acting early on holiday demand is the single biggest revenue lever.

Month-over-Month Revenue Change (%)



Pareto Analysis — Revenue Concentration

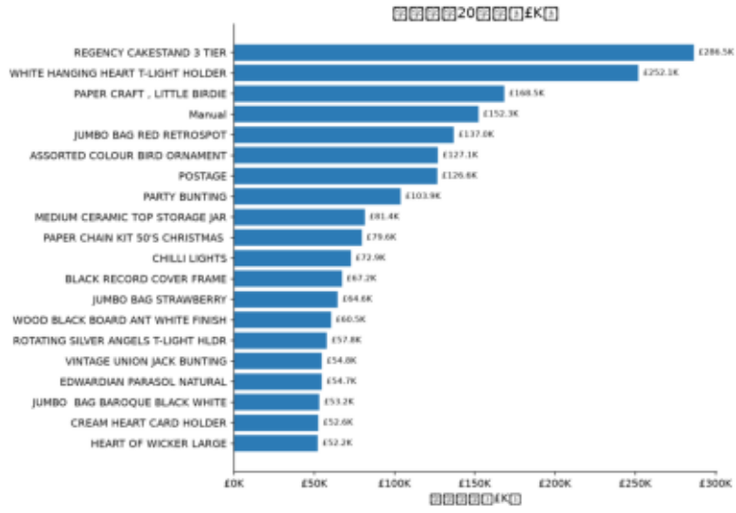


Analysis Commentary

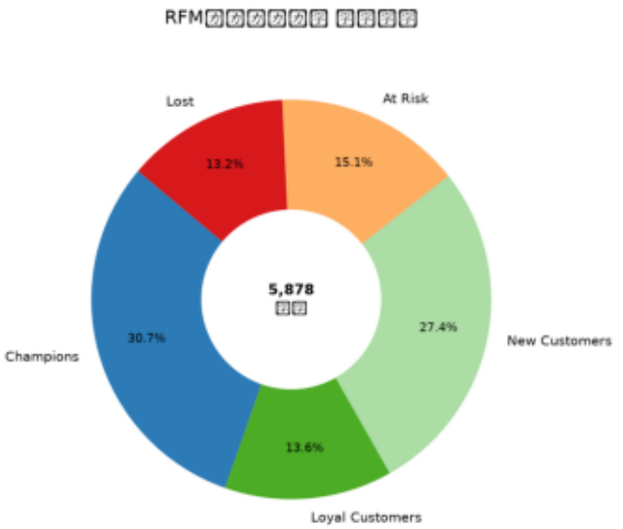
- The Pareto principle holds: the top 20% of SKUs account for the large majority of revenue.
- High-order-frequency products with low revenue share are candidates for a modest price increase.

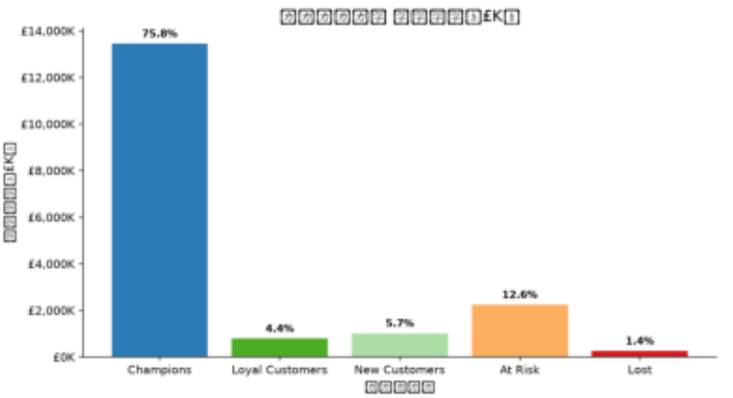
→ Zero stockouts on the top 10 SKUs is the single most impactful supply-chain target — these items alone represent roughly 40% of total revenue.

Top 20 Products by Revenue



Segment Distribution — Customer Count & Revenue Share



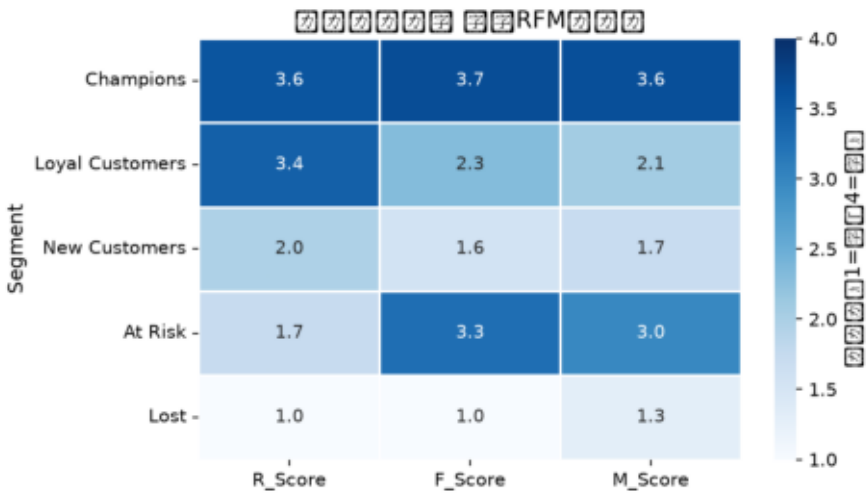


Recommended Actions by Segment

Champions	Recent · Frequent · High-value	VIP perks, early access to new products
Loyal Customers	Repeat buyers, high engagement	Loyalty programme, subscription discounts
At Risk	Past high-value, recently quiet	Limited-time coupon within 30 days
New Customers	First or second purchase	7-day post-purchase follow-up email
Lost	Long-lapsed, no recent activity	Win-back campaign or retire from list

→ Champions are ~31% of customers but drive ~76% of revenue. Protecting this group has a higher ROI than acquiring new customers.

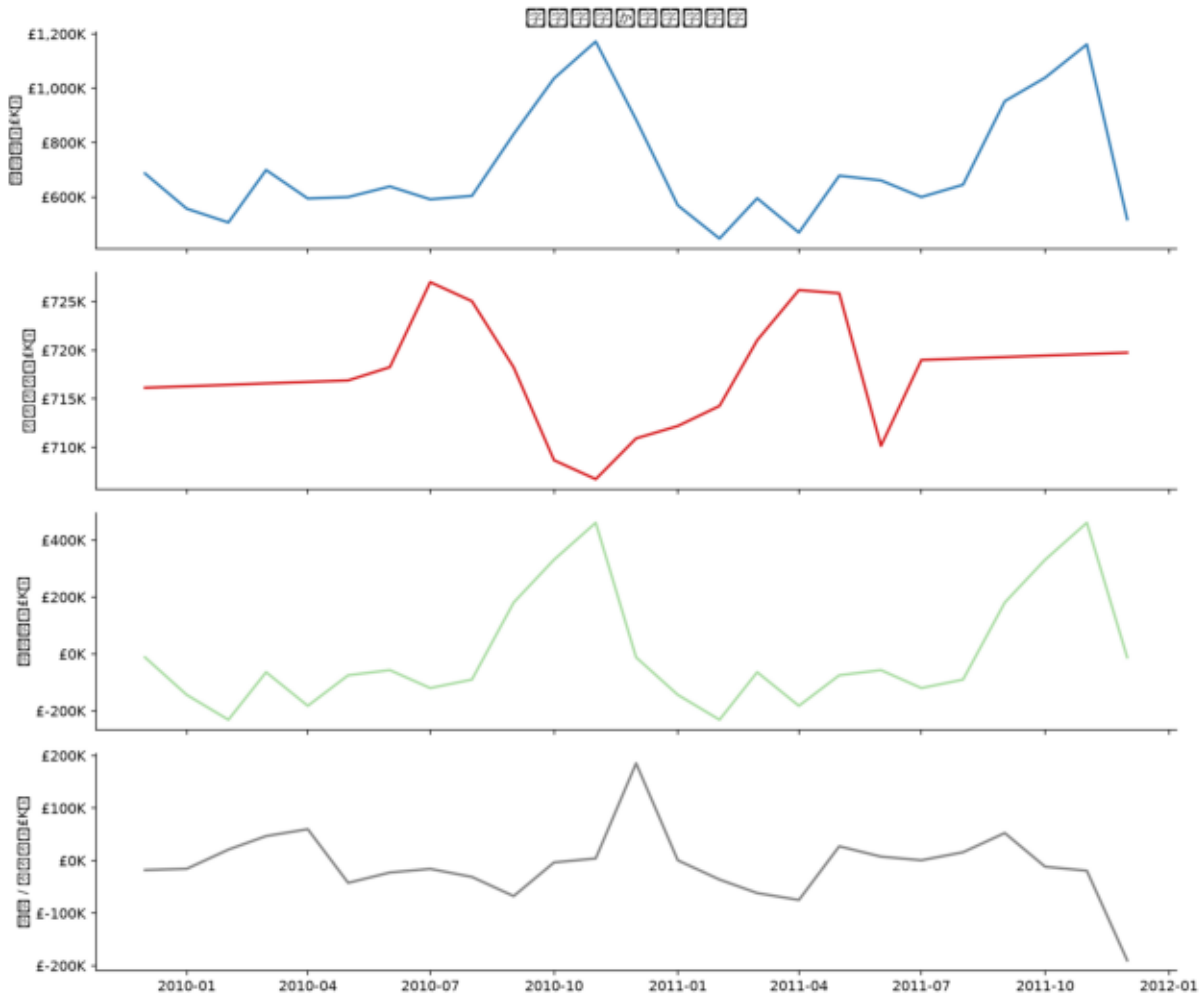
RFM Score Heatmap by Segment



Average R (Recency), F (Frequency), and M (Monetary) scores per segment. Champions score high across all three dimensions; Lost customers score low across all three.

Seasonal Decomposition (Trend · Seasonality · Residual)

Revenue is decomposed into long-term trend, seasonal pattern, and residual noise. The upward trend persists after removing seasonality — confirming genuine underlying growth.



Methodology

This report was produced using industry-standard analytical methods. Results are presented in plain language — no technical background required.

Data Cleaning

Removed cancellations, missing customer IDs, and invalid prices. Only reliable records enter the analysis.

Trend Analysis

Monthly and weekly aggregation with year-on-year comparison. Seasonal decomposition isolates structural growth.

Pareto Analysis

Visualises revenue concentration. Identifies which products and customers deserve the most focus.

RFM Segmentation

Scores customers on Recency, Frequency, and Monetary value, then classifies them into 5 actionable groups.

Enquiries & Engagements

Active data scientist. Available for freelance analysis projects.

- Send your Excel or CSV — that's all we need to get started
- Delivered as a branded PDF or Excel workbook
- Plain-language commentary with every chart
- Happy to scope open-ended or unclear briefs

Get in touch — first conversation is free.

This is a sample report. All analyses are customised to your actual business data and objectives.